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ROTARY DIE CUTTING DEVICE AND METHOD FOR SETTING A GAP DIMENSION OF A GAP BETWEEN A DIE CUTTING CYLINDER AND A COUNTER PRESSURE CYLINDER OF THE ROTARY DIE CUTTING DEVICE

Abstract

A method for setting a gap dimension of a gap between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device includes providing a control system including a controller, at least one sensor, and an adjusting device actuated by the control system. The method further includes actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state, thereby setting the gap dimension between the rotary die cutting device and the counter pressure cylinder, in which actuating the adjusting device is achieved as a result of the control system operating in a closed feedback loop. The method further includes sensing by the at least one sensor occurring with or without disruption of operation of the rotary die cutting device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/217,226 filed on Mar. 30, 2021, the contents of which are incorporated by reference as if fully disclosed herein

FIELD OF THE INVENTION

[0002] The present invention is directed to die cutting apparatus and methods for controlling die cutting apparatus.

BACKGROUND OF THE INVENTION

[0003] Operational & quality control of die cutting processes is of course as old as the invention of the die cutting itself. Control schemes for this process have been in large part manual in nature. Although control schemes involving the use of electrical actuators has recently gained popularity, control schemes using feedback to achieve fully automated control have received little attention. Such control schemes require skilled operators to make process modification determinations, typically requiring shutting down operation of the production line, which is inefficient, disruptive, and costly. Moreover, such control schemes do not include pre-emptive corrective actions encountered during operation of the die cutting apparatus, such as automatically adjusting a gap dimension between a rotary die cutting device and a counter pressure cylinder prior to encountering a splice joint of the material being fed therebetween, the absence of such pre-emptive corrective actions resulting in premature wear of the contact surfaces.

[0004] There is a need in the art for a die cutting apparatus and control schemes that do not suffer from the above shortcomings.

SUMMARY OF THE INVENTION

[0005] In one embodiment, a method for setting a gap dimension of a gap between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device includes providing a control system including a controller, at least one sensor, and an adjusting device actuated by the control system. The method further includes actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state, thereby setting the gap dimension between the rotary die cutting device and the counter pressure cylinder, in which actuating the adjusting device is achieved as a result of the control system operating in a closed feedback loop. The method further includes sensing by the at least one sensor occurring with or without disruption of operation of the rotary die cutting device.

[0006] In one embodiment, the step of actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state includes the deviation from a desired state being a splice joint of at least one of a backing web and an overlying second web.

[0007] In one embodiment, the step of actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state includes the deviation from a desired state being a change in thickness of at least one of a backing web and an overlying second web.

[0008] In one embodiment, the step of actuating the adjusting device in response to at least one sensor of the plurality of sensors sensing a deviation from a desired state includes the deviation

from a desired state being at least one of a change of material of at least one of a backing web and an overlying second web (the overlying second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder), thermal expansion of at least one of the backing web and the overlying second web, a change in humidity of an environment surrounding the rotary die cutting device, a change in a tension gradient of at least one of the backing web and the overlying second web, a cut depth of the punched labels and the matrix web, and an impression quality of the punched labels and the matrix web.

[0009] In one embodiment, the control system employs artificial intelligence methods to control the rotary die cutting device.

[0010] In one embodiment, the control system employs machine learning methods to control the rotary die cutting device.

[0011] In one embodiment, the control system employs deep learning methods to control the rotary die cutting device.

[0012] In one embodiment, a method for setting a gap dimension of a gap between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device includes providing a control system comprising a controller, at least one sensor, and an adjusting device actuated by the control system. The method further includes actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state, thereby setting the gap dimension between the rotary die cutting device and the counter pressure cylinder, in which actuating the adjusting device is achieved as a result of the control system operating in a closed feedback loop. During operation of the rotary die cutting device, a backing web and an overlying second web are directed between the die cutting cylinder and a counter pressure cylinder, the second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder. The method further includes sensing by the at least one sensor occurring in an absence of or prior to separation of the punched labels, the backing web, and the matrix web from one another. The method further includes sensing by the at least one sensor occurring without disruption of operation of the rotary die cutting device.

[0013] In one embodiment, a rotary die cutting device includes a die cutting cylinder, a counter pressure cylinder separated from the die cutting cylinder by a gap, and a control system comprising a controller, at least one sensor, and an adjusting device actuated by the control system. The rotary die cutting device further includes the adjusting device is operably connected to the die cutting cylinder and the counter pressure cylinder. The rotary die cutting device further includes in response to the at least one sensor sensing a deviation from a desired state, the control system is adapted to operate in a closed feedback loop for setting a gap dimension of the gap between the rotary die cutting device and the counter pressure cylinder. The rotary die cutting device further includes during operation of the rotary die cutting device, a backing web and an overlying second web are directed between the die cutting cylinder and the counter pressure cylinder, the second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder. The rotary die cutting device further includes sensing by the at least one sensor occurring in an absence of or prior to separation of the punched labels, the backing web, and the matrix web from one another. The rotary die cutting device further includes sensing by the at least one sensor occurring with or without disruption of operation of the rotary die cutting device.

[0014] In one embodiment, a method for setting a gap dimension of a gap between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device includes providing a control system including a controller, at least one sensor, and an adjusting device actuated by the control system, and components of the control system communicate with one another within a machine network. The method further includes actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state, thereby setting the gap dimension between the

rotary die cutting device and the counter pressure cylinder, in which actuating the adjusting device is achieved as a result of the control system operating in a closed feedback loop. The method further includes sensing by the at least one sensor occurring with or without disruption of operation of the rotary die cutting device.

[0015] In one embodiment, a method for setting a gap dimension of a gap between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device includes providing a control system comprising a controller, at least one sensor, and an adjusting device actuated by the control system, and components of the control system communicate with one another within a machine network. The method further includes actuating the adjusting device in response to the at least one sensor sensing a deviation from a desired state, thereby setting the gap dimension between the rotary die cutting device and the counter pressure cylinder, in which actuating the adjusting device is achieved as a result of the control system operating in a closed feedback loop. During operation of the rotary die cutting device, a backing web and an overlying second web are directed between the die cutting cylinder and a counter pressure cylinder, the second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder. The method further includes sensing by the at least one sensor occurring in an absence of or prior to separation of the punched labels, the backing web, and the matrix web from one another. The method further includes sensing by the at least one sensor occurring without disruption of operation of the rotary die cutting device.

[0016] In one embodiment, a rotary die cutting device includes a die cutting cylinder, a counter pressure cylinder separated from the die cutting cylinder by a gap, and a control system comprising a controller, at least one sensor, and an adjusting device actuated by the control system, and components of the control system communicate with one another within a machine network. The rotary die cutting device further includes the adjusting device is operably connected to the die cutting cylinder and the counter pressure cylinder. The rotary die cutting device further includes in response to the at least one sensor sensing a deviation from a desired state, the control system is adapted to operate in a closed feedback loop for setting a gap dimension of the gap between the rotary die cutting device and the counter pressure cylinder. The rotary die cutting device further includes during operation of the rotary die cutting device, a backing web and an overlying second web are directed between the die cutting cylinder and the counter pressure cylinder, the second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder. The rotary die cutting device further includes sensing by the at least one sensor occurring in an absence of or prior to separation of the punched labels, the backing web, and the matrix web from one another. The rotary die cutting device further includes sensing by the at least one sensor occurring with or without disruption of operation of the rotary die cutting device.

[0017] Other features and advantages of the present invention will be apparent from the following more detailed description, taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 is an upper perspective view of an exemplary rotary die cutting device.

[0019] FIG. 2 is a partial upper perspective view of an exemplary sensor for measuring wear.

[0020] FIG. 3 is an exemplary die strike image guide.

[0021] FIG. 4 is an exemplary die strike image.

[0022] FIG. 5 is an exemplary max pooling of the image of FIG. 4.

[0023] FIG. 6 is a schematic of an exemplary deep learning method how to classify rotary die

cutting device impressions by using convolutional filters to feed imagery grayscale intensity values into a convolutional neural network.

[0024] Wherever possible, the same reference numbers will be used throughout the drawings to represent the same parts.

DETAILED DESCRIPTION OF THE INVENTION

[0025] The invention makes the die cutting process less prone to human error and reduces downtime/costs from damages caused by splices, extends die life & produces higher quality and consistent product, reduces waste by perfecting the impression such that only the waste matrix is lifted off the product, increase throughput by eliminating stoppages to address the aforementioned problems. More specifically, the die depth cut formed in a material passing between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device is automatically controlled, employing a closed feedback loop, utilizing sensors that permit continuous or uninterrupted operation of the rotary die cutting device.

[0026] FIG. 1 shows an exemplary rotary die cutting device **10** including a die cutting cylinder **12** and a counter pressure cylinder **14** separated from the die cutting cylinder **12** by a gap **16**. Stated another way, gap **16** is the distance between the blades of die cutting cylinder **12** and counter pressure cylinder **14**. An adjusting device **18** such as an actuator powered by an electric motor, a hydraulic motor or other suitable power source is operably connected to counter pressure cylinder **14** for increasing or decreasing a dimension or gap dimension of gap **16**, depending upon the direction of travel of adjusting device **18** (i.e., away from or toward die cutting cylinder **12**). In one embodiment, adjusting device **18** may be arranged differently, such as being operatively connected to one or both of die cutting cylinder **12** and counter pressure cylinder **14**. A pair of actuators **20**, **22** such as actuators powered by electric motors, hydraulic motors or other suitable power sources, as controlled by controller **36** via respective conduits **114**, **116** or via wireless communication therebetween, are operably connected to opposite ends of die cutting cylinder **12** for selectively applying forces or pressures (force/area) to the ends of die cutting cylinder **12** for controlling pressure between die cutting cylinder **12** and counter pressure cylinder **14**. More specifically, a backing web **28** and an overlying web **30** are directed between die cutting cylinder **12** and counter pressure cylinder **14**, which overlying web **30** becoming punched labels **32** and a matrix web **34** subsequent to being directed between the die cutting cylinder **12** and the counter pressure cylinder **14**. A pair of sensors **24**, **26** are operably in contact with opposite ends of die cutting cylinder **12** for sensing parameters such as pressure measurements associated with the operation of respective actuators **20**, **22** that are communicated with a controller **36** or processor or control unit. A control system **38** includes controller **36**, actuators **20**, **22**, adjusting device **18**, and sensors **24**, **26**, as well as other sensors to be discussed in additional detail below in the various embodiments. Controller **36** is adapted or configured to provide alerts/logging information **40** and receive historical run data or updates to software or firmware **44** from a storage **42** or storage device such as cloud storage or the internet of things (“IoT”) as is well known and not further discussed herein.

[0027] In one embodiment, sensor **46**, as controlled by controller **36** via conduit **118** or via wireless communication therebetween, measures a thickness of backing web **28**, overlying web **30** or both that remains subsequent to being directed between the die cutting cylinder **12** and the counter pressure cylinder **14**, which thickness measurement may occur subsequent to or prior to removal of matrix web **34**. A database of historical runs (stored in storage **42**) corresponding with the same or similar die cutting cylinder **12**, counter pressure cylinder **14**, overlying web **30** material and thickness is used to lookup what gap **16** was used to produce desirable results in the past. An algorithm in the controller **36** determines the signal to send to adjusting device **18** via conduit **48** or via wireless communication therebetween in order to achieve the same gap **16** that was returned from the database in storage **42**.

[0028] In one embodiment, sensor **50** sends signals into the input nodes of a neural network located in storage **42** via conduit **52** or via wireless communication therebetween. The neural network

classifies the die cut **54** along the perimeter of corresponding punched labels **32** according to predefined categories. The classified die cut from the neural network is compared to a desired result and an error is determined. An algorithm in controller **36** uses the error to determine the signal to send to adjusting device **18** in order to achieve the desired result. In embodiment, sensor **50** may also be used to detect missing punched labels **76** and send alerts from controller **36** to storage **42**.

[0029] For example, from an exemplary die impression image **56** (FIG. 4), an exemplary max pooling **58** (FIG. 5) is convolved, extracting the outer perimeter region **60** (FIG. 5) for use in a neural network **62** (FIG. 6). That is, as shown in FIG. 6, in neural network **62**, information from outer perimeter region **60** (FIG. 5) from max pooling **58** (FIG. 5) is transported and flattened in a one dimensional array **64** that is provided as nodes of an input layer **66** that are assigned different values (weights and biases) for calculations, sometimes referred to as perceptrons, in hidden layers **68** to die impression predictors **70** identified as a light impression predictor **70A**, a medium impression predictor **70B**, and a heavy impression predictor **70C**. The die impression predictors **70A**, **70B**, **70C** of hidden layers **68** correspond to exemplary output layers **72** identified as actuator **20** (FIG. 1) voltage **72A**, actuator **22** (FIG. 1) voltage **72B**, counter pressure cylinder **14** (FIG. 1) drive speed **72C**, predicted die cutter cylinder **12** (FIG. 1) life remaining **72D**, and predicted bearing service date **72E** (associated with one or both of die cutter cylinder **12** (FIG. 1) and counter pressure cylinder **14** (FIG. 1)). As a result, in response to being provided with die impression images **56** (FIG. 4) at predetermined time intervals or intervals of punched labels **32** (FIG. 1) passing sensor **50** (FIG. 1), which may include every punched label **32** or other interval of punched label **32** (FIG. 1), neural network **62** (FIG. 6), control system **38** operates utilizing a closed feedback loop permits continuous operation of die cutter device **10** (FIG. 1), i.e., wherein sensing by the plurality of sensors occurs without disruption of operation of the rotary die cutting device. In one embodiment, sensing by the plurality of sensors occurs with disruption of operation of the rotary die cutting device. In other words, in response to at least one sensor sensing a deviation from a desired state, the control system **38** (FIG. 1) is adapted to operate in a closed feedback loop for setting a gap dimension of the gap **16** (FIG. 1) between the rotary die cutting device **12** (FIG. 1) and the counter pressure cylinder **14** (FIG. 1). Stated another way, die cut depth adjustment is determined by a neural network.

[0030] The neural network **62** permits different weightings in the hidden layer **68** portion to accommodate different preferences of different operators. In other words, such preferences could be changed, corresponding to the working hours of the different operators, and could be performed manually or automatically, such as upon the operator “clocking in” for a work shift.

[0031] In one embodiment, the neural network operates utilizing one or more sensors sensing parameters in an absence of or prior to separation of the punched labels **32** (FIG. 1), the backing web **28** (FIG. 1), and the matrix web **34** (FIG. 1) from one another, and wherein sensing by the plurality of sensors occurring with or without disruption of operation of the rotary die cutting device **10** (FIG. 1), such as will be described in further detail in one of more of the embodiments.

[0032] Feedback to the automated control process may include specifications or measurements pertaining to the die cutting cylinder **12** (FIG. 1), material being cut (e.g., webs **28**, **30** (FIG. 1)) or other process parameters affecting the quality of the die cut **54** (FIG. 1).

[0033] The above closed feedback loop or automated control process is a significant improvement over current manual processes, in which an operator (not shown) utilizing an exemplary die impression image guide **74** (FIG. 3) (image **74A** corresponding to a light die impression, images **74B**, **74C** corresponding to a range of moderate die impressions, and image **74D** corresponding to a heavy die impression), must first disrupt operation of the rotary die cutting device **10** (FIG. 1), review a portion of the webs, determine the deviation from the desired state, make manual adjustment, then restart operation of the rotary die cutting device **10** (FIG. 1), possibly needing to repeat if the operator's deviation determination was incorrect.

[0034] In one embodiment, control system **38** employs artificial intelligence methods to control the rotary die cutting device **10**.

[0035] In one embodiment, control system **38** employs machine learning methods to control rotary die cutting device **10**.

[0036] In one embodiment, control system **38** employs deep learning methods to control rotary die cutting device **10**.

[0037] In one embodiment, as further shown in FIG. **1**, sensor **78** senses and communicates deviations **80** in linear thickness (e.g., prior to encountering deviation **80** or anomaly from a desired state (e.g., desired or nominal thickness or thickness state) such as a splice, in which one or both of backing web **28** and overlying web **30** has an increased thickness compared to a non-splice portion) to controller **36** via conduit **82** or via wireless communication therebetween so that adjusting device **18** will cause counter pressure cylinder **14** to sufficiently increase gap **16** relative to die cutting cylinder **12** to reduce or eliminate die wear and damage. Subsequent to deviation **80** being directed between die cutting cylinder **12** and counter pressure cylinder **14**, gap **16** between die cutting cylinder **12** and counter pressure cylinder **14** is restored to the gap dimension in use immediately prior to encountering deviation **80**.

[0038] It is to be understood that a deviation from a desired state as a result of a change in thickness of at least one of the backing web **28** and overlying web **30** may also be due to thermal expansion of web material or of equipment, a change in humidity of an environment surrounding rotary die cutting device **10**, a change in a tension gradient of at least one of the backing web **28** and overlying web **30**, a cut depth of punched labels **32** and the matrix web **34**, and an impression quality of punched labels **32** and matrix web **34** or other reasons.

[0039] In one embodiment, a deviation from a desired state may also be as a result of at least one of a change of material of at least one of a backing web **28** and an overlying web **30** (the overlying second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder). In other words, if there is a first roll of backing web **28** and an overlying web **30** and a second roll of backing web **28** and an overlying web **30** with the respective materials of at least one of the webs **28**, **30** being different, i.e., having different hardness values, the pressures applied by actuators **20**, **22** (FIG. **1**) to achieve the same impression depth for each roll would be different.

[0040] In one embodiment, sensor **78** senses and communicates deviations **80** in linear thickness (e.g., splices) to controller **36** so that actuators **20**, **22** respond to reduce the pressure between die cutting cylinder **12** and counter pressure cylinder **14** in order to reduce or eliminate wear and damage to die cutting cylinder **12**. Subsequent to deviation **80** being directed between die cutting cylinder **12** and counter pressure cylinder **14**, pressures applied by actuators **20**, **22** is restored to the applied pressures in use immediately prior to encountering deviation **80**.

[0041] In one embodiment, as further shown in FIG. **1**, sensor **78** senses and communicates deviations **80** in linear thickness, combining the functionality of adjusting both gap **16** utilizing adjusting device **18** and pressures utilizing actuators **20**, **22** as previously discussed.

[0042] In one embodiment, sensor **78** senses and communicates deviations **80** in linear thickness, combining the functionality of adjusting both gap **16** utilizing adjusting device **18** and pressures utilizing actuators **20**, **22** as previously discussed, except a sensor **86**, which positioned vertically above webs **28**, **30** (instead of sensor **78** that is laterally positioned relative to webs **28**, **30**) senses and communicates deviations **80** with controller **36** via conduit **84** or wirelessly therebetween.

[0043] In one embodiment, sensor **78** senses and communicates deviations **80** in linear thickness, combining the functionality of adjusting both gap **16** utilizing adjusting device **18** and pressures utilizing actuators **20**, **22** as previously discussed, or a sensor **86** similarly performs the function of sensor **78**, which sensor **86** is positioned vertically above webs **28**, **30** (instead of sensor **78** that is laterally positioned relative to webs **28**, **30**) senses and communicates deviations **80** with controller **36** via conduit **84** or wirelessly therebetween. In this embodiment, deviations **80** are detected

through use of a trained neural network such as similar to neural network **62** (FIG. **6**).

[0044] In one embodiment, die pressure is controlled to a setpoint, versus measured pressures, through use of feedback control. Sensors **24**, **26** sense and communicate pressure measurements and change die pressure as controlled by controller **36** that is associated with the operation of respective actuators **20**, **22** in order to reduce or eliminate the error between the setpoint and measured pressures as sensed and measured by sensors **20**, **22**.

[0045] In one embodiment, any or all activities from the sensors and actuators are sent to cloud storage.

[0046] In one embodiment, such as shown in FIG. **2**, an arrangement for measuring wear of a component of rotary die cutting device is now discussed. As shown, a component **88**, such as a counter pressure cylinder **14** (FIG. **1**) or other component having similar geometric surfaces as described herein includes a wear surface **90** terminating at a step **92** having a step surface **94** that is generally perpendicular to wear surface **90**, and a step surface **96** that is both parallel to and offset from wear surface **90**. A fixed laser source **98** directs a planar laser beam **100** that is parallel to a rotational axis **106** of component **88**. Laser beam portion **108** strikes wear surface **90** and laser beam portion **110** strikes step surface **96**. A camera-based sensor **102** such as DSE26C infrared sensor manufactured by Maxcess Americas is outfitted with an optical filter such that sensing region **104** of sensor **102** only “sees” or senses laser beam **100**. When sensing region **104** of sensor **102** is arranged between 45 degrees and 60 degrees relative to planar laser beam **100** such that sensing region **104** also senses parallel laser beam portions **108**, **110**, control system **38** (FIG. **1**) may be adapted to measure a distance **112** between laser beam portions **108**, **110**, and compare measured distance **112** with the distance between the same surfaces prior to installation of component **88** into the rotary die cutting device, (i.e., when component **88** was new) thereby permitting calculation of the extent of relative wear between wear surface **90** and surface **96**. As a result, this arrangement permits proactive maintenance measures and/or for alerting operators. In one embodiment, the sensors of previously embodiments are multipurposed for use in this embodiment as well as in their original embodiments.

[0047] In one embodiment, information from one or all sensors of one or all embodiments is transmitted to and from storage **42**, including an IoT framework, an onsite network infrastructure or on a local media located in controller **36**.

[0048] In one embodiment, the neural network is deployed with the ability to perform training on demand at the location of deployment or remotely from an offsite location, both occurring either during operation or not.

[0049] In one embodiment, a custom neural network is constructed for isolated or combined parameters present within the control system **38** (FIG. **1**) or process (i.e., customer neural networks for each different type of web material, each different die cutting cylinder, each different counter pressure cylinder, each different line speed setting, etc.).

[0050] In one embodiment, sensors (such as those detecting deviations in linear thickness) communicate within a machine network that is located at a facility (“on site”). This arrangement allows seamless combinations of new and existing technologies present in the facility by using proprietary communication protocol. The machine network may also comprise various sensors from different sites. The communication protocol may also be non-proprietary.

[0051] For purposes herein, the term “machine network” is intended to mean two or more connected devices sharing time sensitive information with predictable and consistent data delivery timing with the understanding that stable closed loop operation being highly dependent on minimizing data delivery timing jitter. Network-based closed loop control is realized using time stamped sensor information representing the actual output being sent via network to the controller for comparison to the desired output to produce a corrective signal at a desired precise time to the adjusting device. Network communication can be realized using any supported signaling method and protocol packet structure as long as all devices are using compatible strategies.

[0052] Surprisingly, results including the deep learning method used for strike impression characterization handles both transparent and opaque materials with equal effectiveness. Another surprise is the high accuracies that are attainable with little training.

[0053] While the invention has been described with reference to one or more embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims. In addition, all numerical values identified in the detailed description shall be interpreted as though the precise and approximate values are both expressly identified.

Claims

1. A method for setting a gap dimension of a gap between a die cutting cylinder and a counter pressure cylinder of a rotary die cutting device, comprising: providing a system including a controller and at least one sensor, for adjusting the gap dimension between the die cutting device and the counter pressure cylinder; setting a desired trained criterion, which is dependent on properties of a web; sensing, via the at least one sensor, at least a die cut along a perimeter of the web being processed from a desired state which includes a set of one or more detectable parameters with quantifiable and measurable variables, wherein the at least die cut along the perimeter of the web is compared to the desired state and a deviation or an error is determined, wherein the deviation or error is used to produce a signal by the at least one sensor to determine a deviation type from the desired trained criterion; and sensing, via the at least one sensor, a required action to achieve a desired result, wherein the required action is used to produce a signal to correct the deviation or error; wherein sensing by the at least one sensor occurring with or without disruption of operation of the rotary die cutting device.
2. The method of claim 1, wherein the deviation from the desired state is a splice joint of at least one of a backing web and an overlying second web.
3. The method of claim 1, wherein the deviation from the desired state is a change in thickness of at least one of a backing web and an overlying second web.
4. The method of claim 1, wherein the deviation from the desired state is at least one of a change of material of at least one of a backing web and an overlying second web, the overlying second web becoming punched labels and a matrix web subsequent to being directed between the die cutting cylinder and the counter pressure cylinder, a thermal expansion of at least one of the backing web and the overlying second web, a change in humidity of an environment surrounding the rotary die cutting device, a change in a tension gradient of at least one of the backing web and the overlying second web, a cut depth of the punched labels and the matrix web, and an impression quality of the punched labels and the matrix web.
5. The method of claim 1, wherein the deviation type is an indication of at least one of a light cut, a heavy cut, a good cut, or a bad cut.
6. The method of claim 1, wherein the required action is moving the die cutting cylinder to generate at least one of a deeper cut or a shallow cut.
7. The method of claim 1, wherein the required action is not moving the die cutting cylinder and maintaining the die cutting cylinder stationary.
8. The method of claim 1, wherein the required action is moving the counter pressure cylinder to generate at least one of a deeper cut or a shallow cut.
9. The method of claim 1, wherein the required action is not moving the counter pressure cylinder and maintaining the counter pressure cylinder stationary.

10. The method of claim 1, wherein the system employs artificial intelligence methods to control the at least one sensor.

11. The method of claim 1, wherein the system employs machine learning methods to control the at least one sensor.

12. The method of claim 1, wherein the system employs deep learning methods to control the at least one sensor.
